

CENG381 Stochastic Processes

Product-Form Martingales and Jensen's Inequality — Practice 1

Q1 (30 points). Let $X_1, X_2, \dots$ be i.i.d. random variables with $E[X_i] = 1$. Define the product-form process $Z_n = X_1 \cdot X_2 \cdot \dots \cdot X_n$ ($Z_0 = 1$).

- a) Prove that $\{Z_n; n \geq 0\}$ is a martingale by computing $E[Z_n | Z_{n-1}, \dots, Z_1]$. Factor out Z_{n-1} and use independence of X_n from the past.
 - b) Let $X_n = 2$ w.p. $1/2$ and $X_n = 0$ w.p. $1/2$ (so $E[X_n] = 1$). Compute $E[Z_n]$ and $P(Z_n > 0)$ for each n . What is Z_n almost surely as $n \rightarrow \infty$? Does the limit $Z_\infty = \lim Z_n$ satisfy $E[Z_\infty] = \lim E[Z_n]$?
 - c) The example in (b) shows a martingale with constant expectation 1 that converges almost surely to 0. Explain: why does this NOT contradict the definition of a martingale? What additional condition (uniform integrability) would guarantee $E[Z_\infty] = \lim E[Z_n]$?
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Q2 (35 points). Jensen's Inequality: if g is a convex function and X has finite mean, then $E[g(X)] \geq g(E[X])$.

- a) Let $g(x) = x^2$. (i) Verify g is convex ($g''(x) \geq 0$). (ii) Apply Jensen: $E[X^2] \geq (E[X])^2$. What familiar quantity is $E[X^2] - (E[X])^2$? What does Jensen say about this quantity?
 - b) Let $g(x) = e^x$ (convex everywhere). Jensen says $E[e^X] \geq e^{E[X]}$. For $X \sim \text{Normal}(\mu, \sigma^2)$: $E[e^X] = e^{\mu + \sigma^2/2}$. Verify Jensen holds: show $e^{\mu + \sigma^2/2} \geq e^\mu$ and explain why equality holds only when $\sigma^2 = 0$.
 - c) Let $g(x) = -\log(x)$ for $x > 0$ (convex since $g''(x) = 1/x^2 > 0$). Let X take values $a_1, a_2, \dots, a_n$ with equal probability $1/n$ each. Apply Jensen to derive the AM–GM inequality: $(a_1 \cdot a_2 \cdot \dots \cdot a_n)^{1/n} \leq (a_1 + a_2 + \dots + a_n) / n$.
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Q3 (35 points). A stochastic process $\{W_n\}$ is a sub-martingale if $E[W_n | W_{n-1}, \dots, W_1] \geq W_{n-1}$ for all $n \geq 2$ (the process tends to increase on average).

- a) Prove: if $\{Z_n\}$ is a martingale and h is a convex function with $E[|h(Z_n)|] < \infty$, then $\{h(Z_n)\}$ is a sub-martingale. Use Jensen's inequality applied to the conditional expectation.
- b) Let $\{S_n\}$ be the symmetric random walk (steps ± 1 w.p. $1/2$, $S_0 = 0$). (i) Show $\{S_n^2\}$ is a sub-martingale by computing $E[S_n^2 | S_{n-1}]$. (ii) Since $\{S_n^2 - n\}$ is a martingale (proved earlier), what is $E[S_n^2]$ for all n ?

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c) Show $\{S_n\}$ is also a sub-martingale by the same Jensen argument ($h(x) = |x|$ is convex). Is $E[S_n]$ an increasing sequence? For large n , the Central Limit Theorem gives $S_n \approx \sqrt{n} \cdot Z$ where $Z \sim N(0,1)$, so $E[S_n] \approx \sqrt{n} \cdot E[|Z|] = \sqrt{2n/\pi}$. Check: is this consistent with $\{S_n\}$ being a sub-martingale?